

subtitle Basic idea about DC to AC converter

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Inverter Lecture 01

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Lecture 01

- ① What is inverter & why is it used
- ② Type of inverters and brief introduction
- ③ 1-ph half bridge VSI inverter
- ④ simulation study
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What is inverter & why is it used

- The conversion of DC power to AC power by means of static power electronics switches.(variation: magnitude, frequency, number of phases).
- As no AC battery is available, giving AC power for AC motor drive/network from a DC source.

Type of inverters and brief introduction

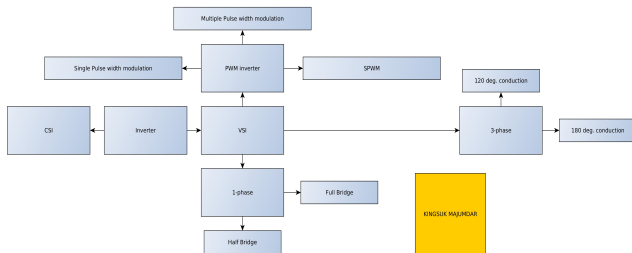


Figure: Type of inverter

1-ph half bridge VSI inverter

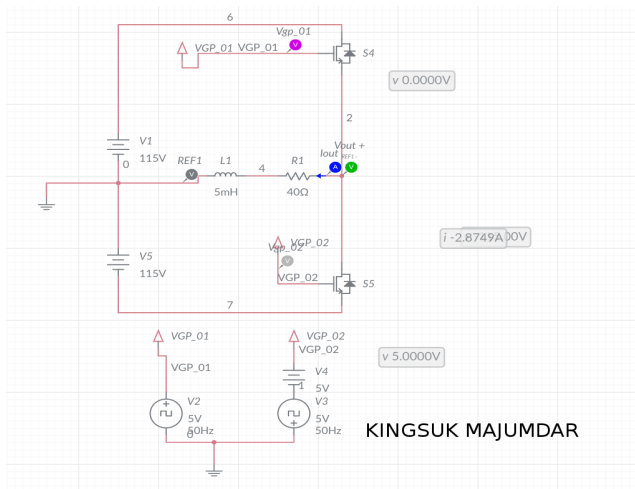


Figure: Half Bridge 1-ph Inverter

cont...

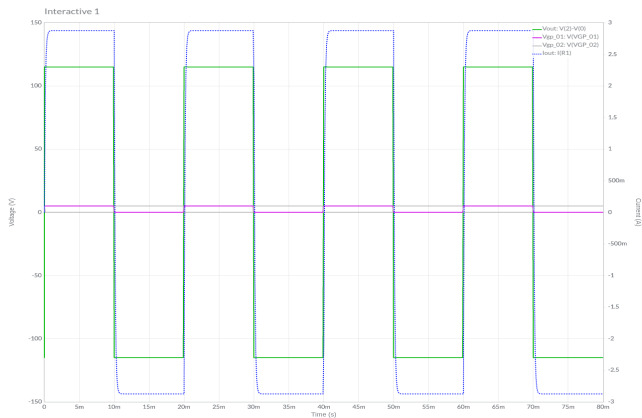


Figure: Voltage waveform

Mathematics of VSI half bridge

$$v_0 = \begin{cases} \frac{V_s}{2}; 0 < t < \frac{T_s}{2} \\ -\frac{V_s}{2}; \frac{T_s}{2} < t < T_s \end{cases} \quad (1)$$

$$V_{0ave} = \frac{V_s}{2} \quad (2)$$

cont...

Again, the instantaneous voltage can be written as

$$v_0 = \sum_{h=1}^{\infty} \frac{2V_s}{h\pi} \sin h\omega t \quad (3)$$

where $h=1,2,3\dots$

NB: $v_s = 0; \forall h = \text{even}$

simulation study

<https://www.multisim.com/content/M9KNZ9MPWePhj4wd4GaLqh/half-bridge-inverter-km/open/>

References



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